**LP - II**

# Mini Project Report

On

## Student Details Manager Using Salesforce Lightning Platform

Submitted in the fulfillment of requirements of Savitribai Phule Pune University

For the degree of **Third Year of Computer Engineering**

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# CERTIFICATE

This is to Certify that -

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On –

Student Details Manager Using Salesforce Lightning Platform

| Prof. Sonal Kadam | Dr. Rahul Patil | Dr. Pramod Patil |
|---|---|---|
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# Acknowledgement

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# Abstract

Cloud computing has transformed the way applications are developed and deployed by providing scalable and on-demand resources over the internet. This project presents the development of a cloud-based application titled **Student Details Manager** using the Salesforce Lightning Platform.

The main objective of this application is to efficiently manage student records in a centralized system. The application allows users to create and manage student information such as Name, Email, Contact Number, Date of Birth, and Branch through custom objects and fields. The user interface is built using Salesforce Lightning, which provides a modern and interactive experience.

By utilizing the Software as a Service (SaaS) model, the system eliminates the need for local infrastructure and enables real-time data access from anywhere. The application ensures better data organization, improved accessibility, and reduced redundancy compared to traditional systems.

This project highlights the practical implementation of cloud computing concepts and demonstrates how platforms like Salesforce can be used to develop secure, scalable, and efficient applications with minimal coding effort.

**Keywords:** Cloud Computing, Salesforce, CRM, Lightning Platform, SaaS, Student Management System

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# CHAPTER 1: PROBLEM STATEMENT

In many educational institutions, student data is still managed using traditional methods such as manual registers or basic offline systems. These approaches are often inefficient, error-prone, and lack proper data organization.

The major problems associated with traditional student data management systems include:

- **Data Redundancy and Inconsistency:** Duplicate entries and lack of synchronization lead to inaccurate records.
- **Limited Accessibility:** Data cannot be accessed remotely, making it difficult for users to retrieve information from different locations.
- **Time-Consuming Operations:** Manual data entry, updating, and searching processes require significant time and effort.
- **Lack of Real-Time Updates:** Changes in student data are not reflected instantly across the system.
- **Poor Data Security:** Traditional systems are more vulnerable to data loss, unauthorized access, and physical damage.

With the growing need for digital transformation, there is a strong requirement for a centralized and cloud-based solution that can efficiently manage student information.

Therefore, this project proposes the development of a **Student Details Manager** using the Salesforce Lightning Platform. The system aims to provide a secure, scalable, and user-friendly platform for storing, accessing, and managing student records in real time, overcoming the limitations of traditional methods.

# CHAPTER 2: SOFTWARE/ HARDWARE REQUIREMENTS

## 2.1 Software Requirements

| Component | Details |
|---|---|
| Platform | Salesforce Lightning |
| OS | Windows / Ubuntu |
| Browser | Google Chrome |
| Account | Salesforce Developer Account |

## 2.2 Hardware Requirements

| Component | Specification |
|---|---|
| Processor | Intel i3 or above |
| RAM | Minimum 4GB |
| Storage | 500MB free space |

# CHAPTER 3: PROPOSED METHODOLOGY

## 3.1 Methodology Overview

The project follows a structured approach starting from requirement analysis to application deployment on the cloud platform. Salesforce Lightning is used to build the application with minimal coding by utilizing its declarative tools.

## 3.2 Steps Followed

1. **Login to Salesforce Developer Account**
   Access the Salesforce environment using a registered developer account.
2. **Creation of Custom Object**
   A custom object named **Student_Detail** is created to store student information.
3. **Field Definition**
   Required fields are added to the object, such as:
   - Name
   - Email
   - Contact Number
   - Date of Birth
   - Branch
4. **Tab Creation**
   A custom tab is created to make the object accessible through the user interface.
5. **Lightning App Development**
   A Lightning Application named **Student Details Manager** is built using App Builder to provide an interactive UI.
6. **Profile and Permission Assignment**
   User roles and permissions are configured to control access to the application and its data.
7. **Data Entry and Testing**
   Sample student records are entered to verify functionality and ensure proper working of the system.

### 3.3 Workflow of the System

- User logs into the Salesforce platform
- User accesses the Student Details Manager app
- User creates or updates student records
- Data is stored in Salesforce cloud database
- Information can be retrieved and managed in real time

### 3.4 Methodology Advantages

- **No Infrastructure Required:** Fully cloud-based system
- **Faster Development:** Uses drag-and-drop tools
- **Scalability:** Can handle increasing data easily
- **Accessibility:** Available anytime, anywhere
- **Security:** Managed by Salesforce cloud platform

# CHAPTER 4: DESIGN

## 4.1 System Architecture

The system follows a cloud-based architecture where all components are hosted and managed on the Salesforce platform.

**Architecture Components:**

- **Cloud Platform:** Salesforce (SaaS model)
- **Frontend:** Lightning User Interface (UI)
- **Backend:** Salesforce Application Server
- **Database:** Salesforce Cloud Database

**Working:**
The user interacts with the Lightning UI through a web browser. The requests are processed by the Salesforce server, and data is stored and retrieved from the cloud database in real time.

## 4.2 Application Design

The application is designed using Salesforce's declarative tools, which allow development without complex coding.

**Main Components:**

- **Custom Object:**
  - *Student_Detail*
- **Fields in Object:**
  - Name
  - Email
  - Contact Number
  - Date of Birth
  - Branch
- **Lightning Application:**

- *Student Details Manager*

- **User Interface:**
  - Built using Lightning App Builder
  - Provides forms for data entry and record viewing

### 4.3 Data Design

The system stores student information in a structured format within the custom object.

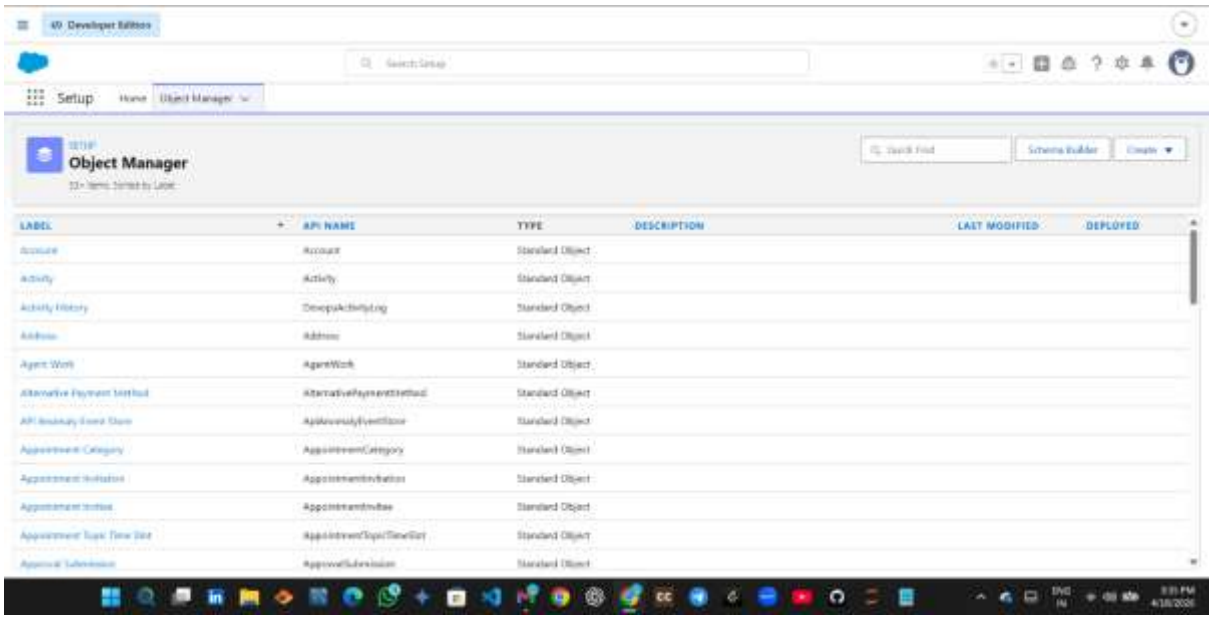
- Each record represents one student
- Fields store specific attributes (e.g., email, contact)
- Data is stored securely in Salesforce cloud storage
- Supports easy retrieval and updates

### 4.4 Design Features

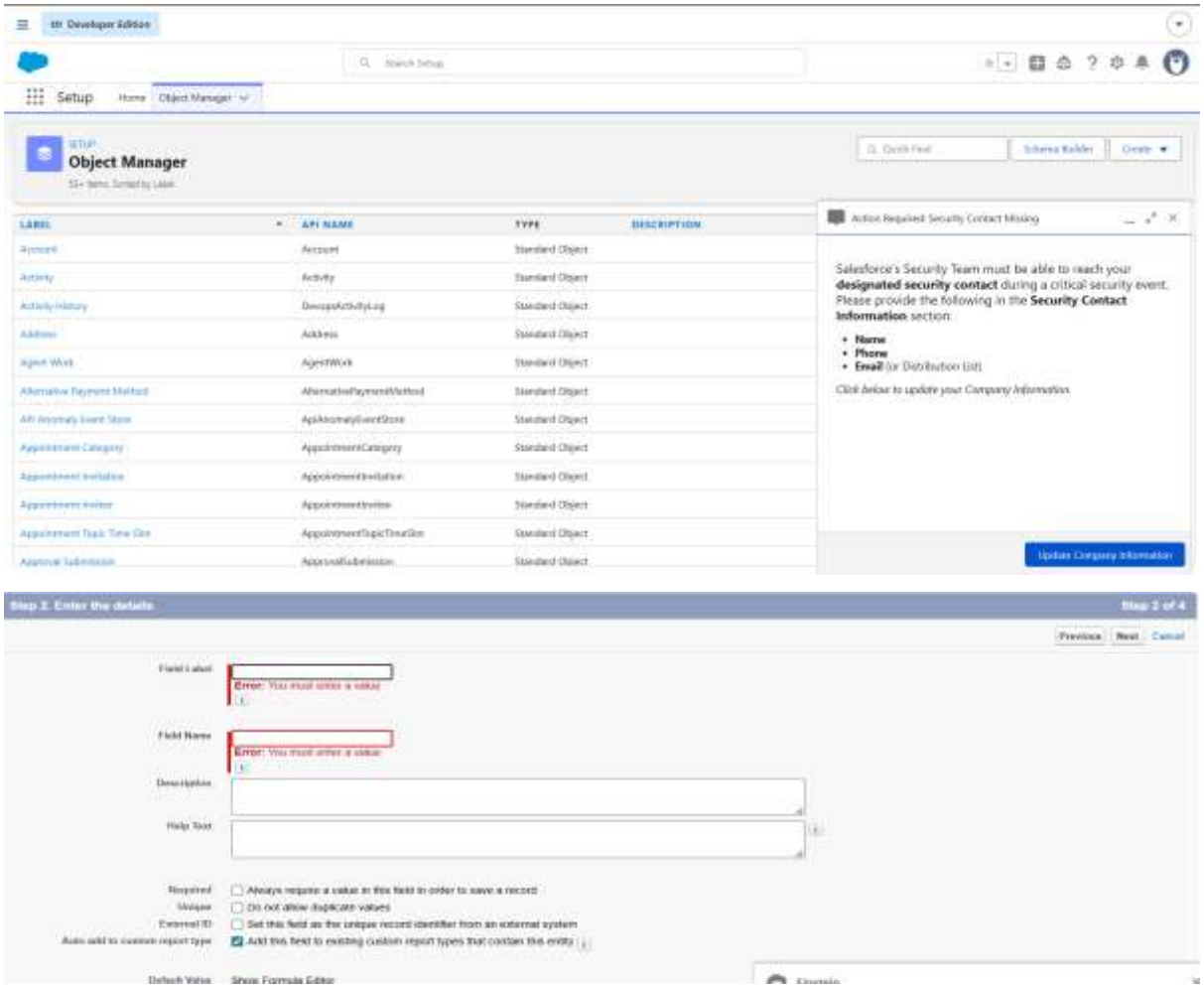
- **User-Friendly Interface:** Simple and intuitive UI
- **Centralized Data Storage:** All data stored in one location
- **Real-Time Access:** Instant data updates and retrieval
- **Scalable Design:** Can handle increasing number of records
- **Secure Architecture:** Data protection managed by Salesforce

# CHAPTER 6: CODE & OUTPUT

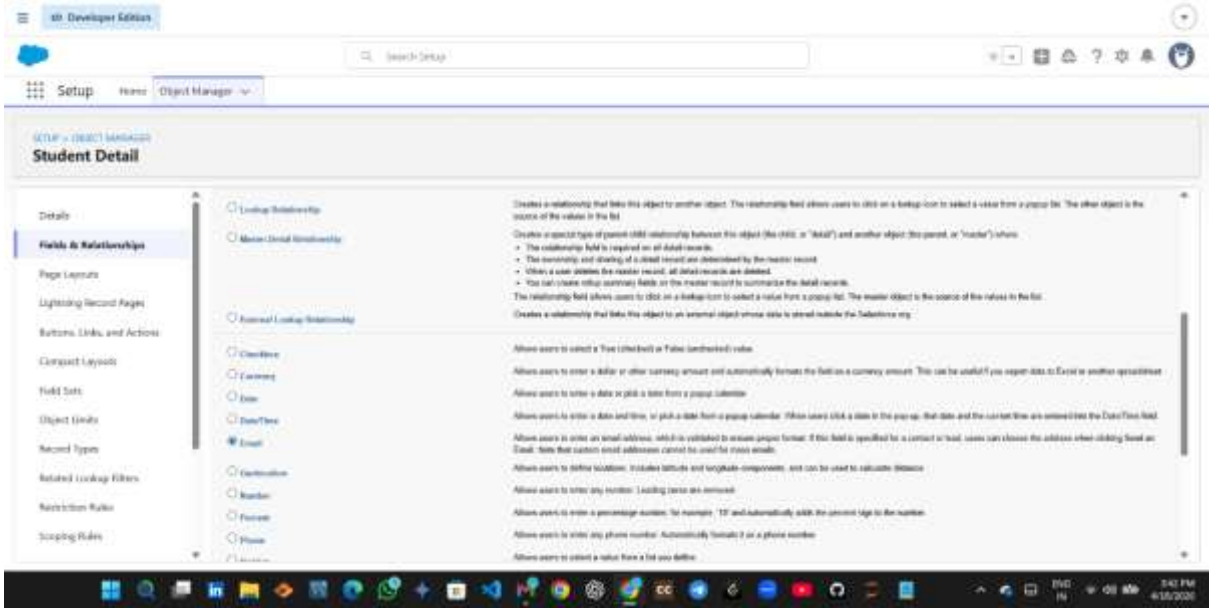
Step 1: Log into Salesforce Developer account & Open Salesforce Lightning platform and click on Object Manager => Create => Custom Object




Step 3: Fill in the required fields and under Optional Features, select Allow Reports and Allow Activities. Click Save

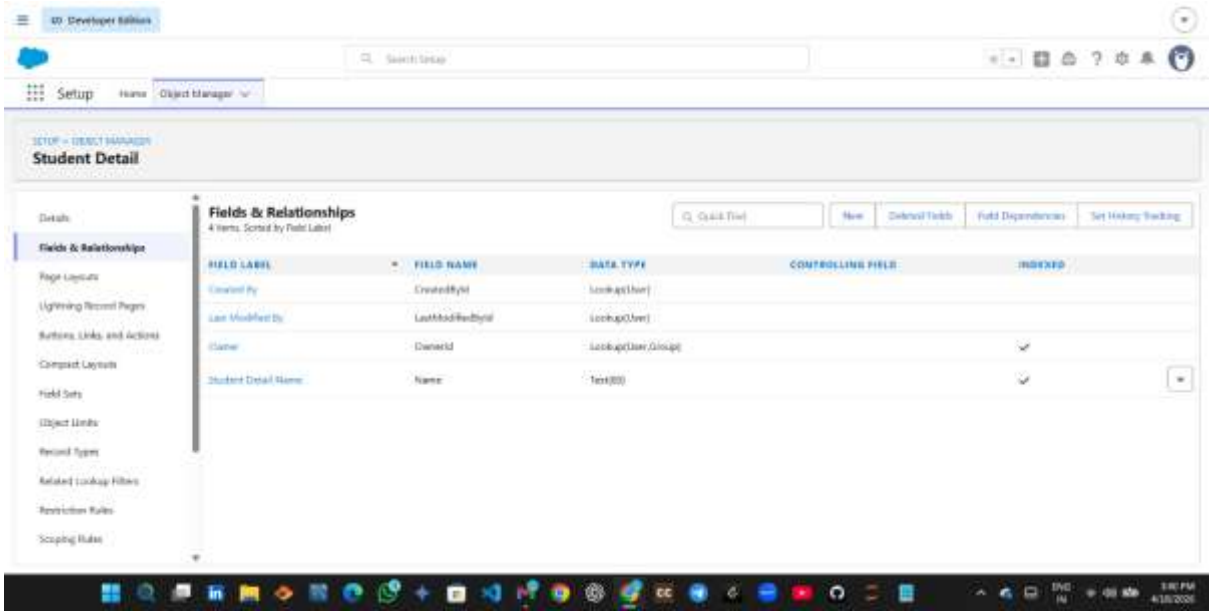


Step 4: Now, Click on Fields & Relations => New.

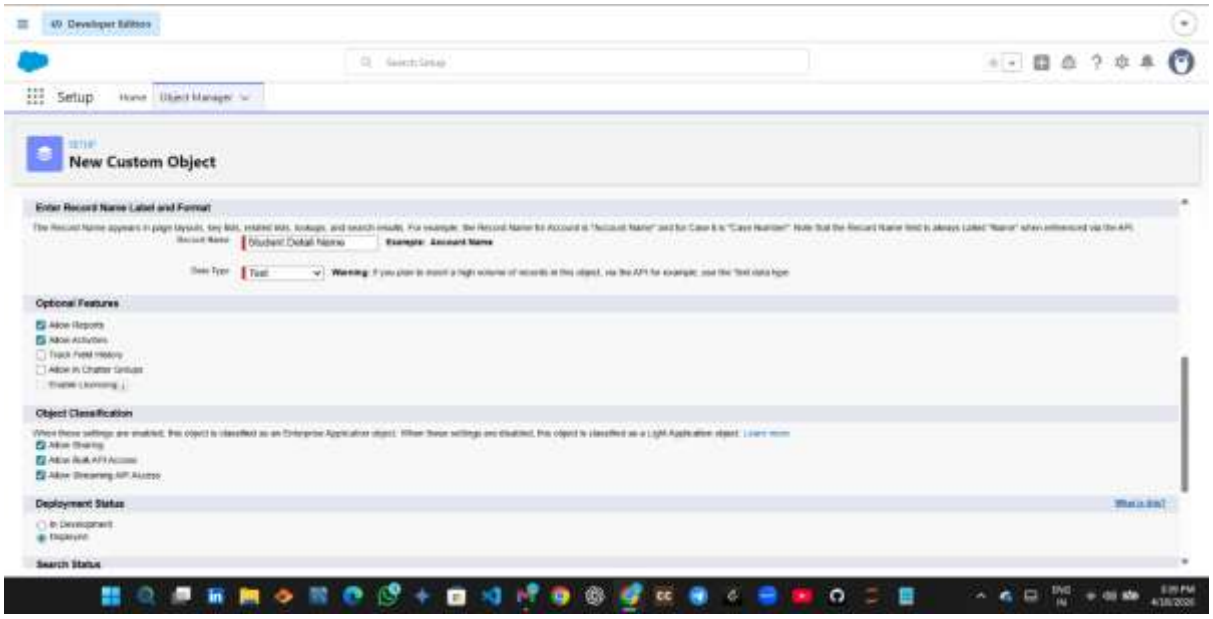



Step 5: Then select option “Email” and click Next-> Next -> Save.

Step 6: Similarly. Repeat steps 4 and 5 to add more fields like Phone, Date of Birth. This is how the custom object will have the various fields.

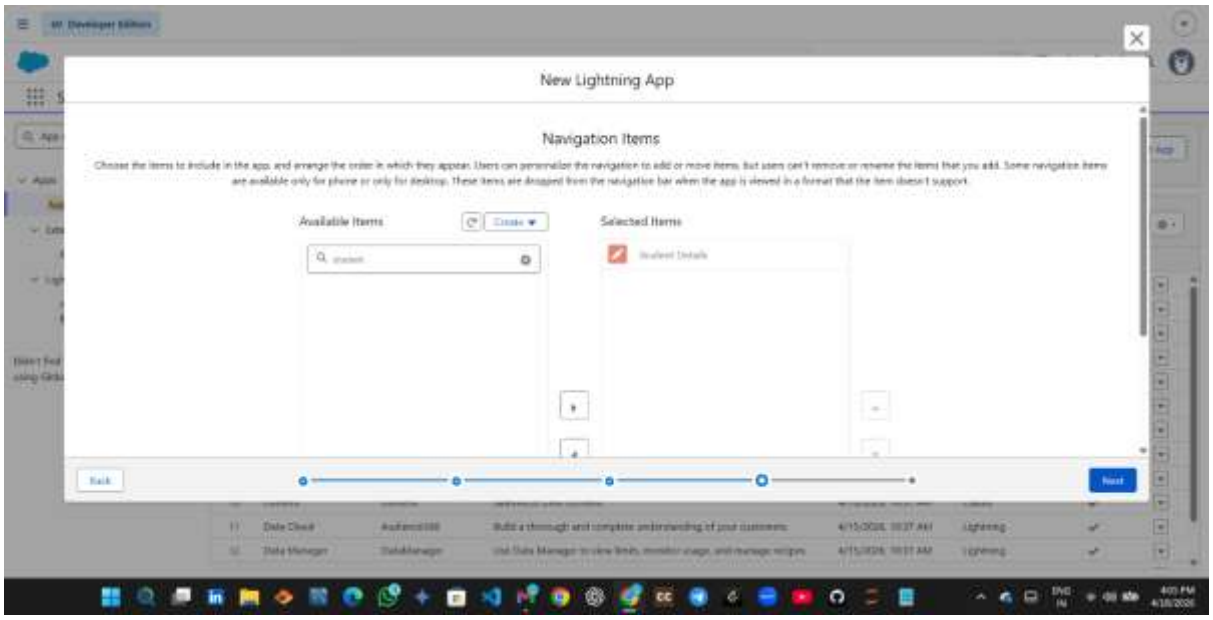



Step 7: Now go to Home => search for “Tabs”. Click on New.




Step 8: Enter the Object name and select any icon for tab style. Leave all defaults as it is. Click Next, Next, and Save.

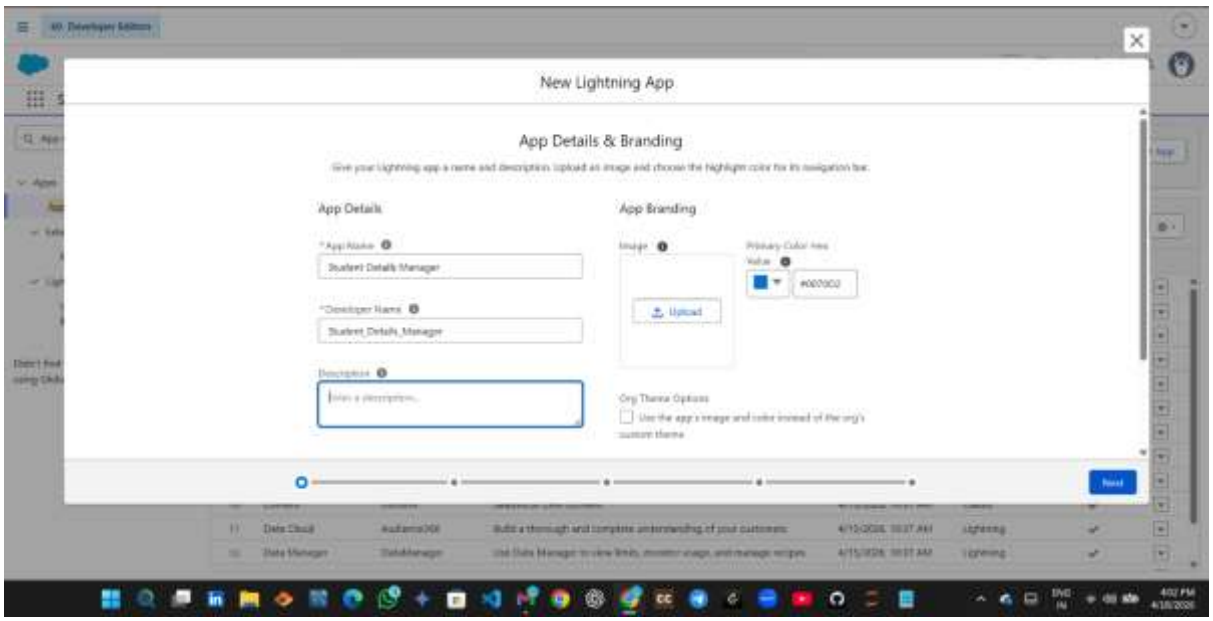
Step 9: In Setup, click Home. Enter “App Manager” in Quick Find and select App Manager. Click New Lightning App.




Step 10: Define the new Lightning app as follows:

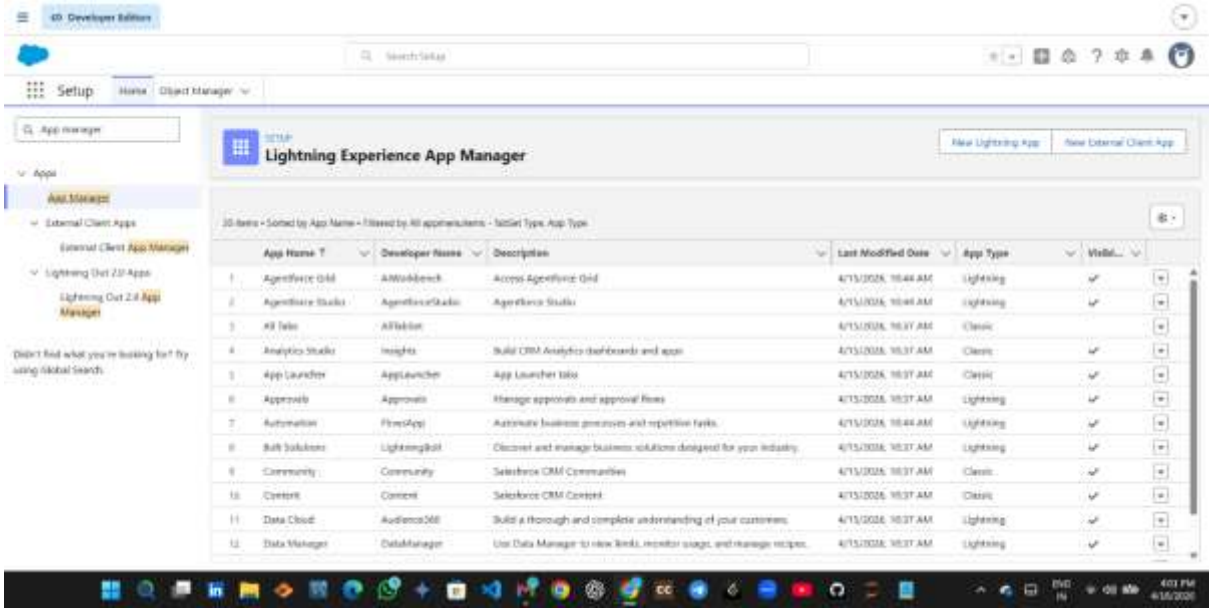
App Name: Student Details Manager

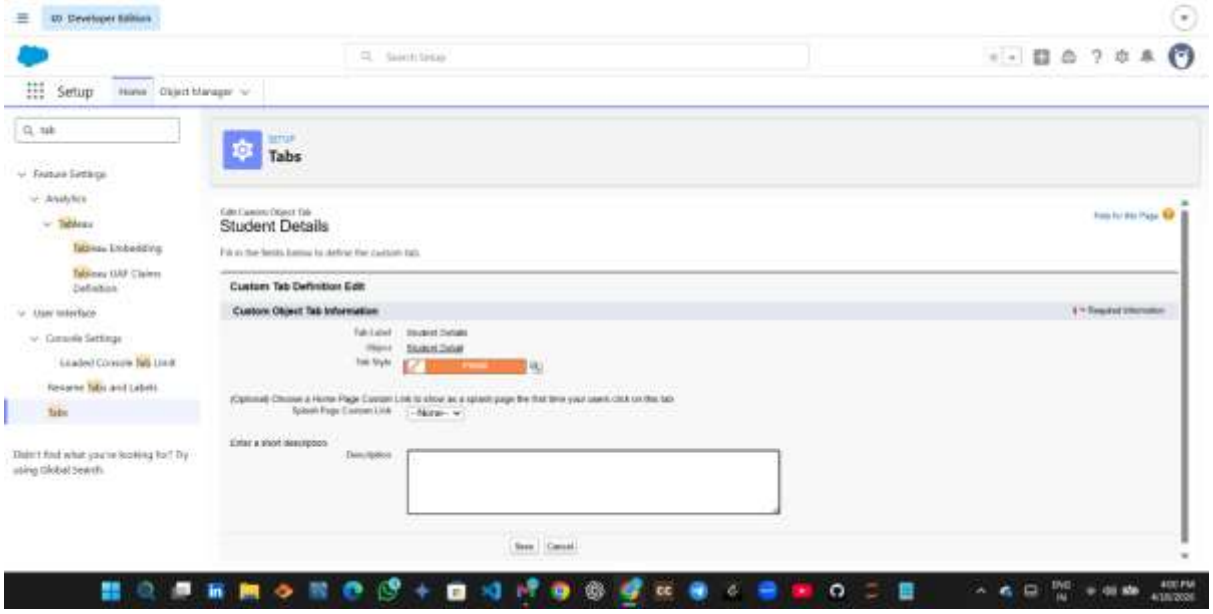
Developer Name: Student_Details_Manager. Click Next



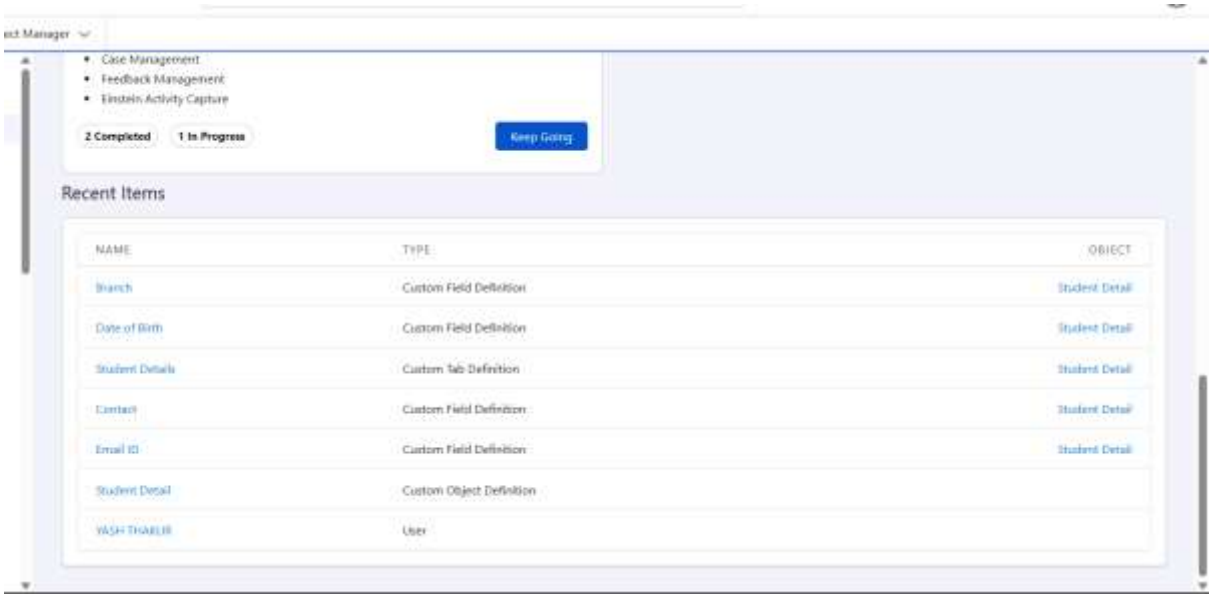

Step 11: On the App Options screen, leave the defaults as is and click Next. On the Utility Items screen, leave the defaults as is and click Next. On the Navigation Items screen, select Student_Detail and move them to the Selected Items box. Then click Next.

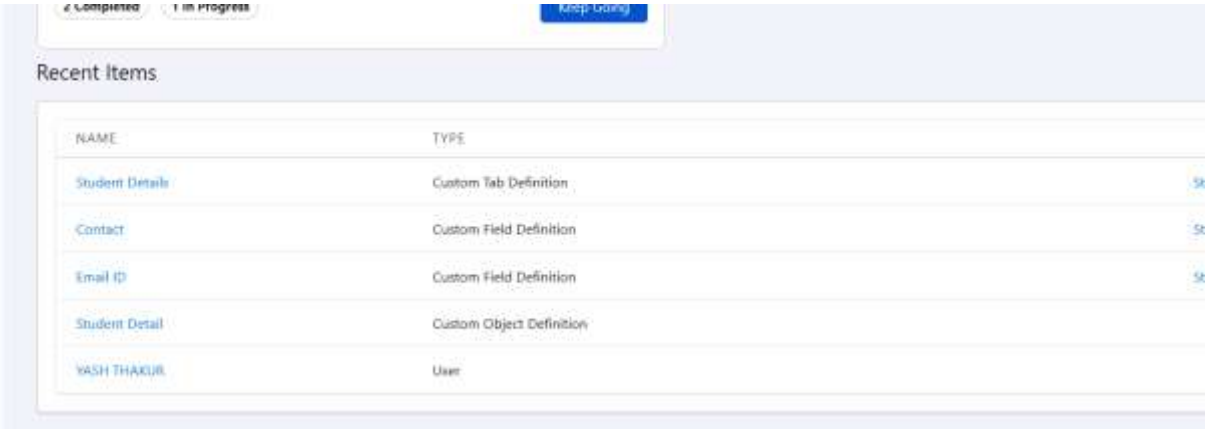
Step 12: On the Assign to User Profiles screen, select System Administrator and move it to Selected Profiles. Then click Save & Finish.

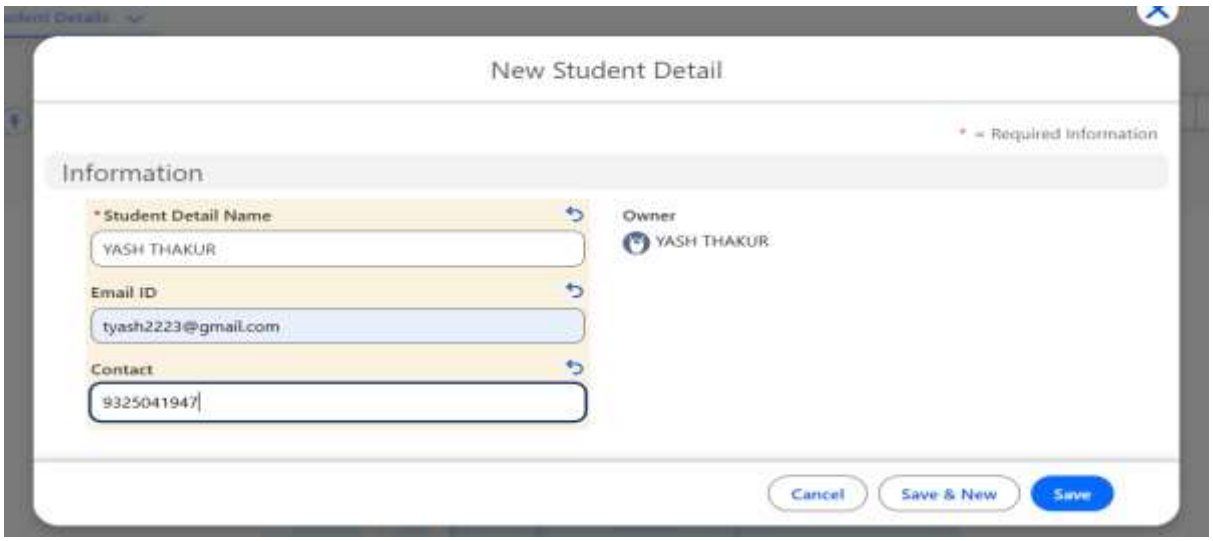


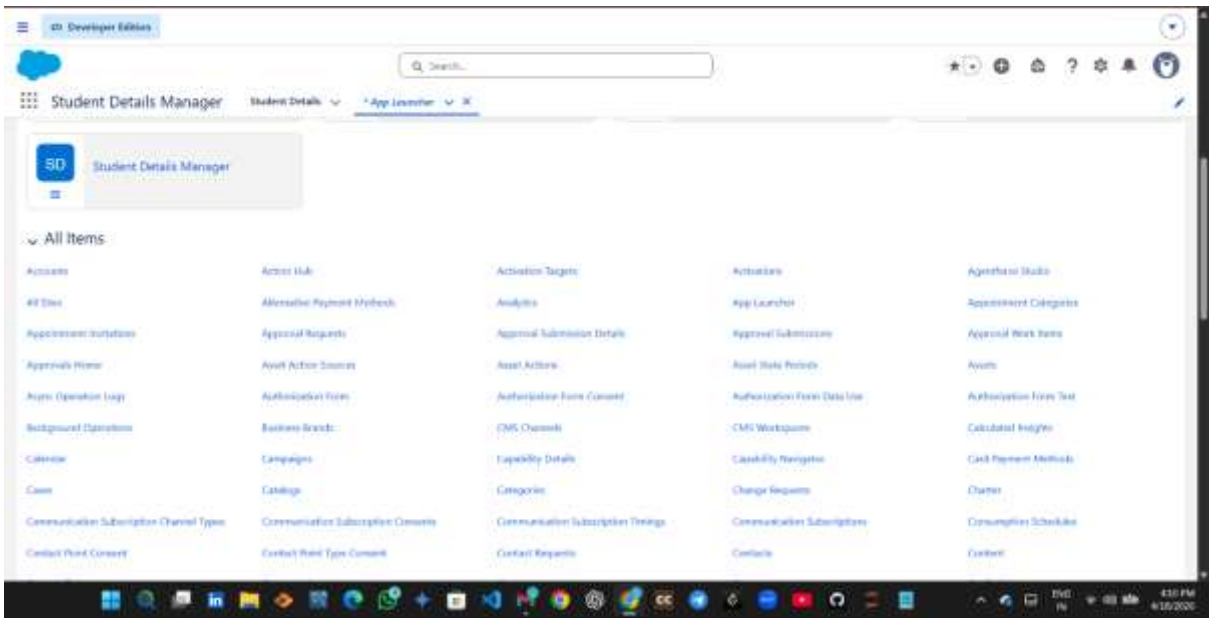



Step 13: Click on App launcher and Open the custom application created

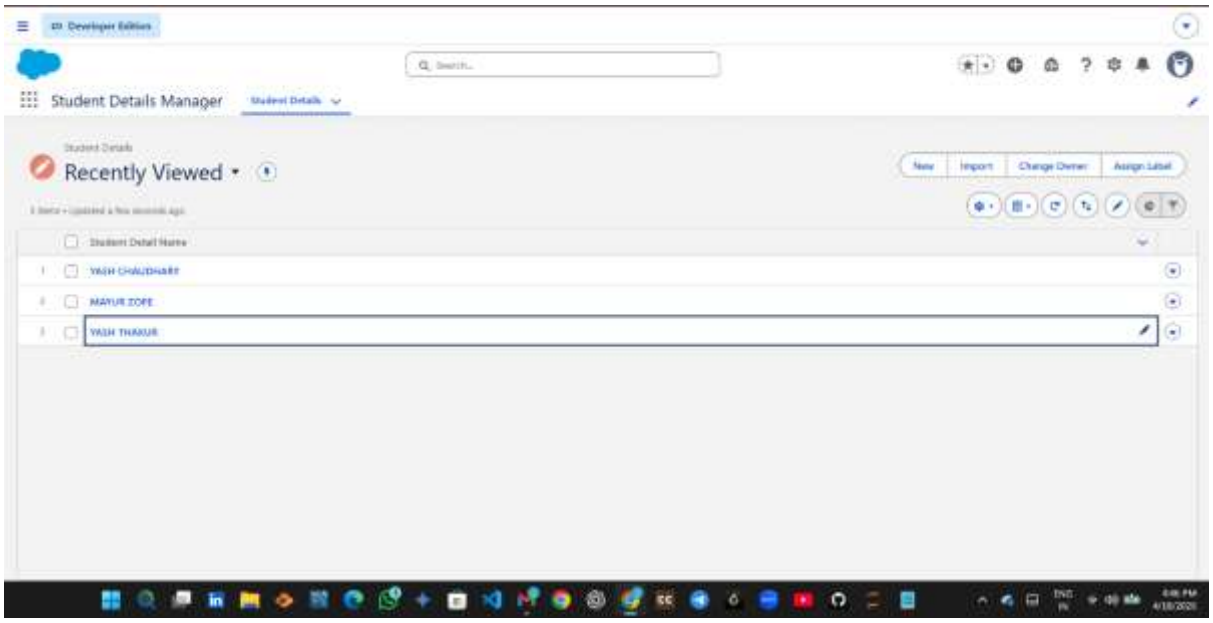


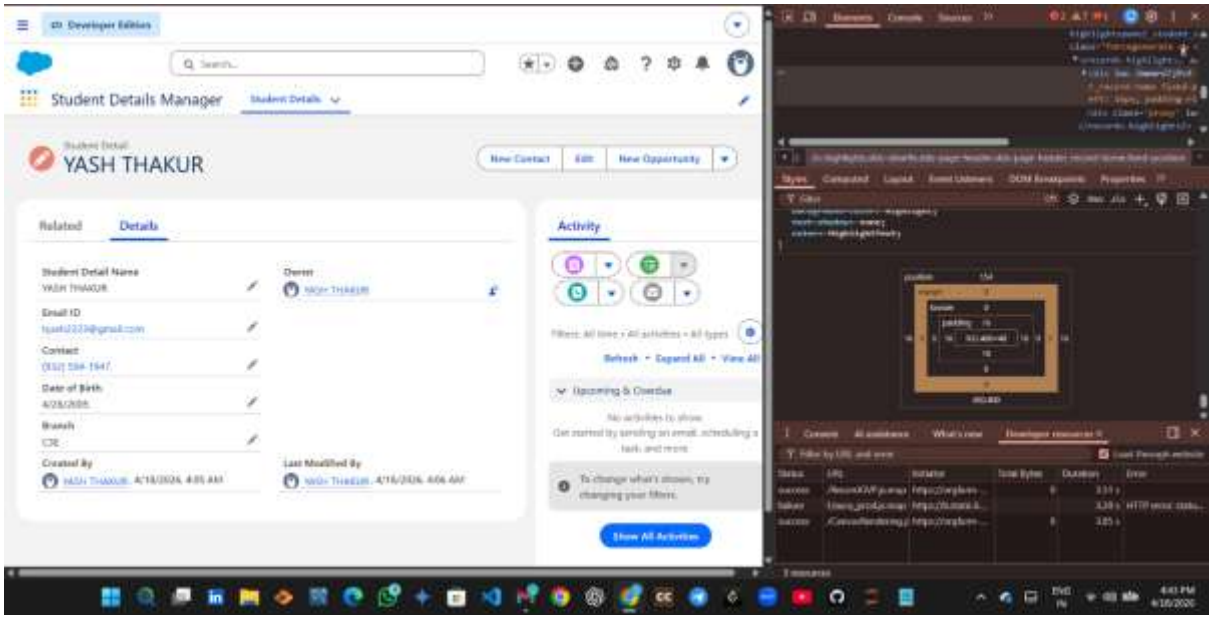


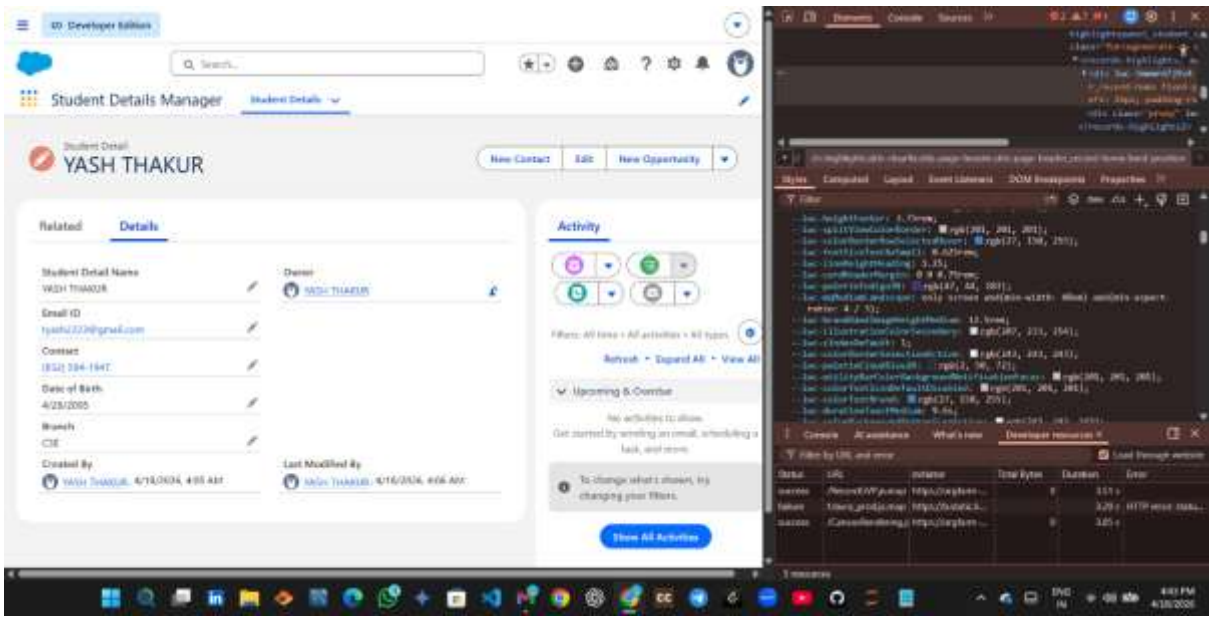




Step 14: Click on Student_Details => New => fill the specified details and copy the URL.



Step 15: Open Google Chrome new Tab => More Tools => Developer Tools and then paste the URL of the application, copied in the previous step



# CHAPTER 7: CONCLUSION

The **Student Details Manager** project successfully demonstrates the development of a cloud-based application using the Salesforce Lightning Platform. The application provides an efficient and centralized system for managing student records with improved accessibility and reliability.

By utilizing cloud computing and the SaaS model, the system eliminates the need for physical infrastructure and enables real-time data access from anywhere. The use of Salesforce Lightning tools simplifies the development process while ensuring scalability and security.

**Key Outcomes**

- Successful creation of a custom object and fields
- Development of a user-friendly Lightning application
- Efficient storage and management of student data in the cloud
- Real-time access and easy data retrieval

**Learning Outcomes**

- Understanding of cloud computing concepts
- Hands-on experience with Salesforce platform
- Knowledge of SaaS-based application development
- Experience in designing and implementing cloud solutions

# CHAPTER 8: REFERENCES

[1] Salesforce Official Documentation https://developer.salesforce.com/docs

[2] Salesforce Trailhead (Salesforce Learning Platform)

[3] Salesforce Lightning Platform Overview

[4] Cloud Computing Concepts and SaaS Model Study Material (College Notes)

[5] Online Tutorials and Technical Blogs on Salesforce Development